

REGRESSION MODEL ASSESSMENT AND MULTICOLLINEARITY

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MATCHING COLOURS, GOOD CORRELATION, BUT A BAD FIT

Selecting an appropriate regression model involves more than simply estimating coefficients. After fitting a model, it is important to assess how well it explains the observed variation in the dependent variable and whether the estimated relationships can be interpreted reliably. Do note however, that a model that appears to fit the data well may still suffer from statistical problems that affect inference. One of the most common of these problems is multicollinearity, which arises when explanatory variables are highly correlated with one another.

MODEL ASSESSMENT AND MEASURING GOODNESS-OF-FIT

2.1 Decomposition of the Sum of Squares

Ordinary Least Squares, OLS, estimators are to minimise the sum of squared residuals, *i.e.*, the sum of squares due to error, SSE . We have the relationship for the sum of squares, $\sum(y_i - \bar{y})^2 = \sum(\hat{y}_i - \bar{y})^2 + \sum(y_i - \hat{y}_i)^2 \Leftrightarrow SST = SSR + SSE$.

SST (Sum of Squares of Total)	$\sum(y_i - \bar{y})^2$	Measures the total variation in y about the sample mean.
SSR (Sum of Squares due to Regression)	$\sum(\hat{y}_i - \bar{y})^2$	Measures the part of total variation that is explained by or due to the regression.
SSE (Sum of Squares due to Error)	$\sum(y_i - \hat{y}_i)^2$	Measures the part of total variation that is not explained by or due to the regression.

2.2 Estimation of Error Variance

We need to estimate the error variance, $\sigma^2 = Var(e_i) = \mathbb{E}[e_i^2]$, but the squared errors are unobservable. Therefore, we develop an estimator for σ^2 based on the squares of the least squares residuals, that is, $\hat{e}_i = y_i - \hat{y}_i = y_i - (b_1 + b_2x_{2i} + \dots + b_Kx_{Ki})$. An estimator for σ^2 that uses the information from $\hat{e}_i$ and has good statistical properties is $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{n-K} = \frac{SSE}{n-K}$, where K is the number of β parameters being estimated in the multiple regression model.

2.3 The Coefficient of Determination

In a multiple regression model, R^2 measures the proportion of variation in the dependent variable explained by all the explanatory variables included in the linear model. The coefficient of determination is $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$. If the model does not contain an intercept parameter, then R^2 is no longer appropriate measure for goodness-of-fit, since it violates the decomposition of the sum of squares, that is, $SST \neq SSR + SSE$. Under these circumstances it does not make sense to talk of the proportion of total variation that is explained by the regression.

2.4 The Adjusted Coefficient of Determination

The problem is that R^2 can be made larger by adding more variables, even if the variables added have no justification. Algebraically, it is a fact that as variables are added, the sum of squared errors, SSE , decreases, and thus, R^2 increases. If, in the extreme case, the model contains $n-1$ variables, then $R^2 = 1$. Thus, an alternative measure of goodness-of-fit is called the adjusted R-squared, denoted by $\bar{R}^2$, calculated as $\bar{R}^2 = 1 - \frac{(\frac{SSE}{n-K})}{(\frac{SST}{n-1})}$. This adjusted measure does not always go up with added variables, because of the degrees of freedom term $n-K$. The problem is that it loses interpretation, $\bar{R}^2$ is no longer the proportion of explained variation. It is usually used as a device for selecting the appropriate set of explanatory variables.

MULTICOLLINEARITY

When looking at data from the result of an uncontrolled experiment, many of the economic variables may move together in systematic ways. Such variables are said to be collinear, and the problem is called collinearity.

Consider the model $y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + e_i$. Thus, $Var(b_2) = \frac{\sigma^2}{(1-r_{23}^2) \sum (x_{2i} - \bar{x}_{2i})^2}$. Exact or extreme collinearity exists when x_{2i} and x_{3i} are perfectly correlated, in which case $r_{23} = 1$ and $Var(b_2) \rightarrow \infty$. Similarly, if x_{2i} exhibits no variation, $\sum (x_{2i} - \bar{x}_{2i})^2 = 0$ and $Var(b_2) \rightarrow \infty$. In this case, x_{2i} is collinear with the constant term. In general, whenever there are one or more exact linear relationships among the explanatory variables, then the condition of exact collinearity exists.

To detect for multicollinearity, the following two methods are commonly used in analysis. One method is to try and fit an auxiliary model on the explanatory variable. If R^2 from this artificial model is high, then the implication is that a large portion of the variation in x_k is explained by variation in the other explanatory variables.

Another method to check for collinearity is by observing the variance inflation factor, VIF, for each regressor. VIF for the k^{th} predictor is $VIF_k = \frac{1}{1-R_k^2}$, where R_k^2 is obtained by regressing the k^{th} predictor on the remaining $K - 2$ predictors. VIF measures how much the variance of the estimated regression coefficient is inflated by the existence of correlation among the predictor variables in the model. A VIF of 1 means that there is no correlation between the predictor and the remaining predictor variables, and thus, the variance of is not inflated at all. The general rule of thumb is that VIFs exceeding 4 warrants further investigation, whilst VIFs exceeding 10 are signs of serious multicollinearity requiring correction. In R, we can use the function `vif()` on the model.

CONCLUSION

The measures discussed in this article help evaluate how successful a model explains variation in the dependent variable, whilst also providing guidance for comparing alternative specifications. Nevertheless, goodness-of-fit measures alone do not guarantee reliable coefficient estimates. Multicollinearity can inflate standard errors and make it difficult to identify the individual effects of explanatory variables, even when the overall model appears to fit the data well. Consequently, both model fit and the relationships among explanatory variables should be examined carefully to ensure that the resulting regression model is both statistically sound and practically useful.

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